Glossary

Technology Years 0–10

Te Tāhuhu o te Mātauranga
Ministry of Education

Word or phrase	Description
Abstraction – see also Black box	The process of simplifying complex systems by exposing only essential features (e.g. a button component in a mobile app) and hiding unnecessary details (e.g. the code that makes the button clickable).
Aesthetics	The visual and sensory qualities of a technological product that influence user experience and desirability, such as a smartphone with a sleek finish or a website with a clean layout and harmonious visuals.
Algorithm	A step-by-step set of rules used to perform tasks or solve problems efficiently in computing, such as a search engine ranking pages or a navigation app calculating routes.
Anthropometric data	Measurements of human body dimensions used to design ergonomically suitable products, such as a fitness tracker shaped to fit various wrist sizes.
Attribute	A characteristic of a technological system, with physical attributes (e.g. size, material) and functional attributes (e.g. behaviour, such as the portability of a tablet being supported by lightweight aluminium body and screen size).
Authentic	Involves tasks or tools that reflect real-world technology practices to deepen understanding, such as building a prototype using sustainable composite materials to explore the trade-offs between strength, weight, and environmental impact in product design.
Behaviour – see also Inputs and Outputs	The observable actions or responses of systems, devices, materials, or software under specific conditions. Behaviour may be programmed, emergent, or user-driven and helps describe how technology interacts with inputs and environments.
Binary	A number system in computing based on two states, 0 and 1, used to represent all digital data and logic.
Black box – see also Abstraction	A component whose internal workings are hidden, interacted with only through its inputs and outputs, such as a software module that processes data without exposing its code.
Boolean values	The two possible values for a Boolean data type, usually expressed as ‘true’ and ‘false’.

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Component	A physical part of an electrical or electronic system that performs a specific function, such as a resistor controlling current or a capacitor storing energy. Components work together to enable the system to operate.
Computational thinking problem	A problem that is solved by breaking it into logical steps that a computer or person can follow. It involves techniques such as decomposition, pattern recognition, abstraction, and algorithm design.
Computer-aided design (CAD)	The use of specialised software to create precise technical drawings and models, such as engineers simulating a product before manufacturing.
Condition	A logical statement in computing that evaluates to true or false to control program flow, such as checking if a user is logged in before showing a dashboard.
Constraint	A limitation or condition that influences the design or development of a technological outcome, such as time, cost, resources, or material properties. Constraints guide decision-making and ensure solutions are practical and achievable.
Control	The ability to direct how a system behaves using inputs or rules, such as a smart light turning on when motion is detected.
Convergent thinking	A problem-solving approach that narrows multiple ideas to one best solution, such as selecting the final design for a product after testing different concepts against user needs, materials, and function.
Current (electricity)	The movement of electric charge through a material (conductor), measured in amperes (A).
Debugging	The process of identifying, analysing, and fixing errors or faults in a digital system, program, or code. It is an essential part of development that ensures the final product functions correctly and meets its intended purpose.
Decisions	In computing, decisions within algorithms use control structures to choose actions based on conditions, enabling dynamic responses to different inputs.
Decomposition	The process of breaking down a complex problem into smaller parts.
Design	The purposeful and creative process of developing solutions that meet specific needs by combining function, form, and user experience.
Design brief	Guiding information outlining the purpose, users, required attributes, and (in later years) specifications of a technology outcome.
Design framework	A semi-structured model that helps students work through a project by identifying a problem or need, exploring ideas, making and testing a solution, and reflecting on how well it works. It

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	supports clear thinking, smart decision-making, and creative design.
Design principles	Foundational guidelines that ensure digital products are functional, consistent, and user-friendly, such as using clear navigation to improve usability.
Design thinking	A problem-solving approach focused on user needs, creativity, and iterative testing, such as developing a prototype through empathy and experimentation.
Digital system	A set of components that process, store, and transmit data using binary code, such as hardware and software working together to perform computations.
Disinformation	False or misleading content deliberately spread to deceive, such as fake news on social media affecting public opinion.
Disposal	The process of safely discarding or recycling technology materials, such as responsibly recycling old smartphones to reduce environmental impact.
Electronic (system)	A set of components using electrical signals to perform functions like data processing or communication, such as a sensor system with input, process, and output stages.
Embedded control systems	Systems that are built into larger products or environments and use sensors, programmable logic, feedback, and outputs to automatically monitor and control system behaviour.
Ergonomics	The study of designing systems to fit human needs and capabilities, such as shaping a keyboard for comfort and reduced strain.
Ethics	Principles guiding right and wrong actions, such as considering privacy and fairness when designing a social media platform.
Feedback	Information that shows how a product or system performs and helps improve its design, usability, or efficiency. It can come from people, such as users or testers, or from automated sources like systems, sensors, or analytics. In technology learning, understanding feedback supports better decisions and continuous improvement. For example, a student adjusts the grip of a handle after users report discomfort, or a smart thermostat logs temperature changes to show if heating cycles are efficient.
Fit-for-purpose	Describes a product, system, or solution that effectively meets its intended use, requirements, and conditions. In technology learning, it encourages students to evaluate whether their designs or outcomes are suitable and practical for the task at hand.
Form and function	The relationship between how a product looks (form – subjective) and how it works (function – objective).
Functional modelling	The creation of simplified representations to show how a system works, such as a diagram of energy flow in a solar-powered device.

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Innovation	The development of new ideas or products that improve or solve problems in original ways, such as designing a wearable device that tracks hydration levels.
Input and Output – see also Behaviour	Input is what goes into a system – like data, information, energy, or materials – and output is what comes out after the system transforms it. For example, a computer takes in data and transforms it into information, a lightbulb or LED takes in electrical energy and transforms it into light, and a toaster takes in bread and transforms it into toast.
Lifecycle	Stages a product goes through, from creation to disposal, and how long it remains useful, such as a phone designed for five years of use before recycling.
Loop	A programming structure that repeats instructions until a condition is met, such as a game character moving until it hits an obstacle.
Malinformation	Information that is based on reality, but is deliberately used to cause harm, often by sharing it out of context or with malicious intent.
Mechatronic system	An integrated system that combines mechanical, electronic, computer, and control engineering to perform complex tasks automatically, demonstrating how hardware and software interact to create intelligent machines.
Microcontroller	A small programmable chip that controls specific functions in electronic devices, such as managing sensor input in a smart thermostat.
Misinformation	False or inaccurate content spread without intent to deceive, such as sharing outdated health advice without verifying its source.
Modelling	The process of creating representations to explore how a product looks or works, such as building a cardboard prototype to test structure or simulate function.
Performance properties	How well a system or material functions under specific conditions. For materials, it depends on what they are made of (composition) and how their parts are arranged (structure), affecting strength, flexibility, or durability. For digital systems, performance depends on certain conditions, such as processing speed or network bandwidth in a cloud service.
Planning tool	A resource or method used to organise ideas, tasks, or processes before beginning a technological project, helping learners structure thinking, set goals, and identify steps needed to achieve a successful and completed outcome.
Processed outcome	An outcome refined through physical, chemical, or biological means, such as turning raw milk into pasteurised dairy products.
Product – see also Outcome	A tangible or digital outcome created to meet user needs, such as a software app developed to solve a scheduling problem.

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Programming	Methods for writing code, such as using block-based drag-and-drop blocks or typing Python commands to build a game.
Pseudocode	A plain-language way of writing programming steps without strict syntax, such as outlining an algorithm using basic logic before coding in a programming language.
Reasoning – see also Decisions	In technology learning, the ability to think through problems and make decisions based on logic and evidence.
Resistance	A measure of how much a material opposes the flow of electric current; measured in ohms (Ω).
Selection	The process of choosing between different options or paths based on specific conditions, often using decision-making structures.
Sensory profiling	Analysing the sensory qualities (e.g. taste, smell, feel) of a food technology outcome to assess its fitness for purpose.
Sequence	A structured progression of learning activities or concepts that build upon foundational ideas and each other to develop understanding and skills.
Smart materials	Materials that respond to changes in their environment – such as temperature, pressure, light, or electricity – by altering their properties in a controlled and useful way.
Specifications	Detailed descriptions of the requirements, features, and constraints that a technological outcome must meet, guiding the development process by clearly outlining what a product or system should do and how it should perform. Specifications are objective and are part of a brief.
Stakeholder	Anyone who has an interest in the outcome and impact of a project, product, or process but has no decision-making authority over the development. Stakeholders can be categorised into various groups, such as end users, clients, community members, and experts.
Structural system	A set of interconnected components designed to support loads and maintain stability within a product or structure.
Structure	The way parts of a product, system, space, or concept are arranged and connected to function effectively.
Subsystem	A smaller, self-contained part of a larger system that performs a specific function within the whole.
Symbols	Special signs or characters used to represent ideas, actions, or objects in a clear and simple way, helping communication of complex information quickly.
System	A set of interconnected components, which can be natural or human-made, that work together, transforming inputs into outputs to perform a specific function or achieve a goal and to solve problems or improve processes.

Word or phrase	Description
Systems thinking	An analytical approach that focuses on understanding how parts of a system interact, how systems evolve over time, and how to design improved systems, considering relationships, feedback loops, and the broader context when solving simple and complex problems.
Technological practice	The process of generating ideas, designing, developing, and implementing technological outcomes. This involves actions like prototyping, testing, evaluating, and applying technical knowledge and skills to address needs or problems, ensuring outcomes are sustainable, socially acceptable, and fit-for-purpose.
Test (testing)	Testing materials, components, resources, or languages to determine which is the most appropriate for a feasible or tangible outcome, such as testing different fabrics for durability before selecting one for a wearable device or running unit tests on code to check if functions work as intended.
Trialling	The process of determining the most appropriate techniques for the development of a feasible or tangible outcome, such as trialling various materials joining techniques or trialling coding frameworks to find the most efficient for building an app.
Transformation	The process of changing or converting something from one form, state, or condition to another.
Usability principles	Guidelines that help ensure a product or system is easy, efficient, and satisfying to use. They focus on aspects like clarity, consistency, feedback, and user control to improve the overall user experience.
Voltage	The potential difference that drives electric current; measured in volts (V).